

QUANTITATIVE PRECIPITATION FORECASTING BY USING CONVOLUTIONAL NEURAL NETWORK AND LONG SHORT-TERM MEMORY HYBRID MODEL

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ABSTRACT. *In recent years, the importance of quantitative precipitation forecasting has increased in flood control and water resource management. This study aimed to improve the accuracy of quantitative precipitation forecasting. The proposed method is a hybrid deep learning model that combines a Long Short-Term Memory (LSTM) network with a Convolutional Neural Network (CNN). Although conventional LSTM models are effective for learning temporal patterns, they cannot capture the spatial relationships among observation stations. In contrast, the proposed model introduces spatial features by utilizing CNN to learn the location information of observation points. The contributions of this study are as follows. First, spatial information is incorporated using CNN, enabling the model to capture spatial dependencies among observation stations. Second, experiments using observational data from 13 stations in Gunma Prefecture confirmed that the model improved 1-hour-ahead quantitative precipitation forecasting accuracy. Third, the evaluation across multiple forecast lead times (1-6 h) shows that the proposed method performs best for 1-hour-ahead predictions. The CNN+LSTM hybrid model achieved better performance on average than the standalone LSTM model in terms of Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Threat Score (TS), particularly for stations surrounded by densely located neighbors. These results suggest that the introduction of spatial information is effective in improving the accuracy of short-term quantitative precipitation forecasting.*

Keywords: Precipitation, Rainfall prediction, LSTM, CNN, Deep learning, Machine learning

1. Introduction. Quantitative precipitation forecasting is important for preventing floods and landslides from occurring. Quantitative precipitation forecasting is also used for water resource utilization. In recent years, an increasing number of studies have incorporated precipitation forecasting into specific cases [1, 2]. Sikder et al. have proposed a technique that rapidly updates flood forecast maps using quantitative precipitation forecasting [1]. Ceppi et al. have used quantitative precipitation forecasting for irrigation [2]. This helps manage the amount of water used.

Data assimilation is a major method for quantitative precipitation forecasting. Miyoshi et al. have performed a 30-minute advance prediction [3]. They have used satellite observation data with a data assimilation approach. Kita et al. have reported an improvement in precipitation forecast accuracy by assimilating X-band MP radar data using a three-dimensional variational method [4]. Improving quantitative precipitation forecasting using data assimilation and numerical models is still important. However, this method requires spatiotemporally abundant observational data from satellites and radars. Therefore, it is difficult to apply this approach to quantitative precipitation forecasting in nonurban areas. Alternative quantitative precipitation forecasting methods are important.

Deep learning has been applied in many fields [5]. This trend has also been observed in weather forecasting [6]. Research in this area is advancing steadily. Matsuoka et al. have used deep learning to predict cloud clusters [6]. They have forecasted cumulonimbus clouds before the formation of tropical cyclones. Suzuki et al. have applied a Convolutional Neural Network (CNN) to predicting heavy rainfall [7]. Deep learning has also been applied to spatiotemporal forecasting problems beyond meteorology. For example, taxi demand prediction has been studied using spatiotemporal graph attention networks [8]. In addition, seq2seq models have been used to generate continuous multi-step forecasts for hydrological time-series predictions. For example, Xiang et al. have applied an LSTM-based seq2seq model to hourly rainfall-runoff forecasting and evaluated it using bias and RMSE-related metrics [9].

Deep learning methods have also been applied to quantitative precipitation forecasting [10]. Recent studies have increasingly focused on spatiotemporal deep learning models. Radar echo maps have been used for short-term rainfall prediction with spatiotemporal recurrent networks enhanced by attention mechanisms [11]. Beyond image-based inputs, recent studies have proposed models that explicitly represent the complex spatial interactions. Graph Neural Networks (GNNs) have been introduced to learn spatial dependencies and couple multiple physical factors for precipitation forecasting [12]. In addition, station-graph-based Spatiotemporal Graph Neural Networks (ST-GNNs) have been proposed for rainfall forecasting using meteorological station networks [13]. Transformer-based architectures have also been applied to rainfall prediction using observation-derived atmospheric information such as GNSS Precipitable Water Vapor (PWV) [14]. These studies indicate that explicitly incorporating spatial information is important for improving the precipitation forecasting performance.

Kaneko and Nakayoshi have proposed a method to predict precipitation one hour later in Kyushu, Japan, by learning data from 114 sites in Kyushu, Japan. This method adapts the Long Short-Term Memory (LSTM) [15]. This architecture is specialized for learning time-series data. This study has demonstrated the potential of the LSTM model for precipitation prediction. However, this poses a challenge. LSTM-based quantitative precipitation forecasting cannot learn the spatial locations of the observation stations. Therefore, it may fail to capture the spatial relationships between the stations. If the model learns both the time series and station locations, the prediction accuracy may improve.

This study aims to improve the accuracy of quantitative precipitation forecasting using LSTM. However, a key limitation of LSTM is its inability to consider the spatial locations of the observation stations. To address this issue, we propose a method that combines LSTM with CNN. A CNN is used to extract the coordinate data from each station. This spatial information helps the LSTM to learn the station locations. In the proposed method, each observation station is mapped to a fixed grid based on its latitude and longitude. A CNN extracts spatial features from the grid-based representation. These features are then fed into LSTM layers for temporal modeling. As a result, the model can

learn the station location information and spatial relationships among nearby stations, which a conventional LSTM cannot capture. Specifically, we build a quantitative precipitation forecasting model that predicts rainfall one hour ahead. It uses observational data from the past 12 h as input. We evaluate the proposed model by comparing its accuracy with that of a standard LSTM model. In addition, it is important to understand how the prediction accuracy changes with longer forecast horizons. Therefore, this study also evaluates the performance of the proposed model at multiple forecast lead times of 1, 2, 3, 4, 5, and 6 h. The remainder of this thesis is organized as follows. Section 2 describes the observational data used in this study. Section 3 explains the structure of the proposed CNN+LSTM hybrid model. Section 4 introduces the evaluation metrics adopted to evaluate the models' performance. Section 5 presents the experimental results and analyses. Finally, Section 6 concludes the paper and discusses the future work.

2. Data. Data are obtained from the Japan Meteorological Agency website [16] for 13 observation stations in Gunma Prefecture, Japan. The Japan Meteorological Agency provides observational data for three types of stations. The first type is a four-element station. These stations measure the temperature, precipitation, wind speed, and sunshine duration. The second type is rainfall stations that measure precipitation. These stations record only precipitation. The third type is meteorological station. It observes temperature, precipitation, wind speed, sunshine duration, pressure, and relative humidity. In this study, we limit the number of stations from which to obtain data to “meteorological station” and “four elements stations” in Gunma Prefecture, as shown in Figure 1 [17, 18]. Data are obtained from 13 stations. Hourly data from each observation station are used for training. We use data from January 1, 1996, 1:00 AM to January 1, 2022, 0:00 AM for training. In addition, data from 1:00 AM on January 1, 2022, to 0:00 AM on January 16, 2025, are used as test data.

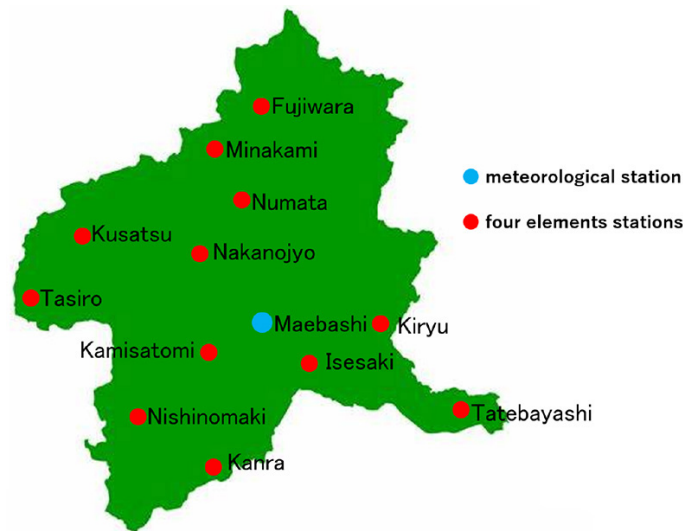


FIGURE 1. Observation points in Gunma Prefecture

The training data include temperature, precipitation, and horizontal wind speed components. Wind direction and wind speed data are obtained from the Japan Meteorological Agency. It is provided in two components: wind direction and wind speed. We convert these into two components: the east-west component wind speed u and the north-south component wind speed v .

3. **Model.** In this case, we use the LSTM model, which predicts precipitation for the next hour based on 12 h of observational data. Each dataset contains input from the previous 12 h at each station. The target output is a quantitative precipitation forecasting for one hour later. The model is evaluated for one-hour-ahead forecasts as well as for 2, 3, 4, 5, and 6 h ahead. In each case, the model uses observational data from the previous 12 h as input. The target output is set to the corresponding future time.

Because there are missing measurements in the observation data, the values are interpolated using linear interpolation according to the number of missing measurements. As proposed in [15], the screening procedure is as follows.

- 1) If there are missing temperature data, linear interpolation is performed if the missing data are within 4 hours in a row.
- 2) If precipitation and wind speed are absent, linear interpolation is performed if the absence is within 1 hour continuously.

For missing value processing, we adopt the same method as Kaneko and Nakayoshi [15]. Linear interpolation is applied only to short gaps in the observation records. Specifically, temperature data are interpolated when the missing period is within four consecutive hours, whereas precipitation and wind data are interpolated only when the missing period is within one hour. A one-hour gap is the minimum possible gap in our hourly records, and restricting interpolation to this case reduces the risk of creating spurious rainfall events. Some missing values may remain even after the screening. If any of the observed data for each feature for the past 12 h and the precipitation data for one hour later to be forecast contain missing measurements, that dataset is not used.

To maintain computational stability during training, standardization is performed for all features to reduce them to a distribution of values with a mean of 0 and a variance of 1. The standardized value w is calculated by

$$w = \frac{x - x'}{\sigma}, \quad (1)$$

where x is the value before standardization, x' is the mean value, and σ is the standard deviation [15].

We use Keras (3.8.0) and TensorFlow (2.18.0) to build the LSTM model. The model has three LSTM layers. These are combined and output as the final output. The first and second LSTM layers output time-series representations, and the final LSTM layer outputs a fixed-length representation for the precipitation prediction. A dropout is applied after each LSTM layer to reducing overfitting.

We describe a CNN that encodes the spatial arrangement of the stations. Each observation station is placed on a fixed grid based on its latitude and longitude. We first determine the latitude and longitude ranges from the station coordinates. We then draw grid lines that are evenly spaced within these ranges. Each station is assigned to the nearest grid cell. If a grid cell contains no station, the feature values of that cell are set to zero. This grid representation allows the model to use station location information. Each grid cell (row, col) stores the time series of the features for the station assigned to that cell.

The data pass through a CNN and are flattened into a 1D vector. This vector is input into the first LSTM layer. The setup of the CNN is described as follows. First, a 2D convolution layer (Conv2D) is applied at each time step to extracting spatial features. The ReLU function is used as the activation function to perform nonlinear transformations. By setting the padding to SAME, convolution is performed while maintaining the input size and edge information. Next, a pooling layer (MaxPooling2D) is applied. This layer extracts important features while reducing the size of the input data. In the final step,

the data are converted into a 1D vector. This prepares the data in a form that is suitable for the LSTM layer.

4. Evaluation Metrics. We use the Root Mean Square Error (RMSE), Mean Absolute Error (MAE), BIAS, and the Threat Score (TS) to evaluate the forecast performance of the models. These metrics are used for comparisons between the models.

The RMSE is expressed by

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - x'_i)^2}, \quad (2)$$

where N is the total number of data, x_i is the predicted precipitation, and x'_i is the measured precipitation.

The MAE is expressed by

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |x_i - x'_i|, \quad (3)$$

The MAE represents the average magnitude of the prediction errors without considering their direction.

The BIAS is expressed by

$$\text{BIAS} = \frac{1}{N} \sum_{i=1}^N (x_i - x'_i), \quad (4)$$

BIAS indicates whether the model systematically overestimates or underestimates precipitation.

The threshold value of precipitation is defined as a [mm/h]. If precipitation exceeds a [mm/h], the case is labeled “with precipitation”. Otherwise, it is labeled “without precipitation”. The number of cases where precipitation is evaluated as “with precipitation” but there is no actual precipitation is FX, the number of cases where precipitation is evaluated as “without precipitation” but there is precipitation is XO, and the number of cases where precipitation is evaluated as “with precipitation” but there is actual precipitation is FO. TS is expressed by

$$\text{TS} = \frac{\text{FO}}{\text{FX} + \text{FO} + \text{XO}}. \quad (5)$$

5. Results. In the experiments, we compare our forecasts with two simple baseline methods: persistence and moving-average methods. In the persistence method, the predicted precipitation at the future time step is set to be equal to the most recent observed precipitation. This method is equivalent to the naive forecasting method [19]. In the moving-average method, the prediction is calculated as the average precipitation over the past k hours, where k denotes the window length [19]. As an additional reference model, we also include a Spatiotemporal Graph Neural Network (ST-GNN), which explicitly models the spatial dependencies among observation stations [13]. In this method, each station is treated as a node in the graph. Edges are constructed using a k -nearest-neighbor graph based on geographical coordinates. At each time step, the node features consist of temperature, wind, and precipitation. Spatial features are extracted using a Graph Convolutional Network (GCN), followed by a residual GCN block and a Graph Attention Network (GAT) layer. The target station embeddings over the previous 12 h are then fed into a bidirectional Gated Recurrent Unit (GRU) to model the temporal dependencies. A fully connected

layer outputs the precipitation forecasts for each target station. An ST-GNN is introduced to provide a graph-based reference for comparison with the proposed CNN+LSTM model.

The spatial grid resolution has been set to six rows and six columns. The grid interval has been approximately 0.15 degrees in latitude and 0.21 degrees in longitude. In this grid mapping, no multiple observation stations have been assigned to the same grid cell. This choice balances the spatial resolution and computational cost, given the limited number of observation stations. The input sequence length has been set to 12 h. We have evaluated forecast lead times from 1 to 6 h ahead to examine how the accuracy changes with longer horizons. For the CNN component, we have used 32 filters to extract the spatial features from the grid-based representation. This setting has maintained a lightweight model and has helped avoid GPU memory issues. In preliminary trials, increasing the number of filters substantially has increased the computational cost and has made the training unstable, whereas the resulting accuracy gain has been limited. For temporal modeling, three LSTM layers have been set to 512, 256, and 256 units. This configuration has balanced the model capacity and generalization performance. Smaller hidden sizes have reduced forecasting accuracy. In contrast, larger hidden sizes have tended to cause unstable training and overfitting issues. The dropout rates have been set to 20% and 50% for the first and second and third LSTM layers, respectively. A dropout has been applied after each LSTM layer to suppressing overfitting. Higher dropout rates have been used in the deeper layers, where overfitting has been more likely to occur. The first and second LSTM layers have used a many-to-many setting and output time-series representations. The third LSTM layer has used a many-to-one setting and output a fixed-length vector. We have used the Adam optimizer with a learning rate of 0.001. The loss function has been defined as the mean squared error. The batch size has been set to 128 samples. The model has been trained for 50 epochs using the Adam optimizer. For TS, the threshold value of precipitation has been set to $a = 1$ [mm/h].

A comparison of the model performance based on the RMSE and TS is summarized in Table 1. Table 1 shows that the average RMSE of the CNN+LSTM hybrid model is 0.012 [mm/h] lower than that of the standalone LSTM model. It also shows that the average TS of the CNN+LSTM hybrid model is 0.006 higher. In addition, Table 2 provides a comparison using the MAE and BIAS. Table 2 shows that the average MAE of the CNN+LSTM hybrid model is 0.021 [mm/h] lower than that of the standalone LSTM model. These results indicate that the combination of CNN improves the prediction accuracy. Regarding BIAS, both models exhibit a small negative bias on average, indicating that the predicted precipitation tends to be slightly lower than the observed precipitation. Furthermore, Table 1 shows that the persistence and moving-average methods yield higher mean RMSE and lower mean TS than the CNN+LSTM hybrid model. Similarly, Table 2 shows that the persistence and moving-average methods yield higher mean MAE than the CNN+LSTM hybrid model. These results indicate that the proposed CNN+LSTM hybrid model provides a better forecasting performance than simple reference methods.

TABLE 1. Comparison of model performance by RMSE and TS

	LSTM		CNN+LSTM		Persistence		Moving-Average		ST-GNN	
	RMSE	TS	RMSE	TS	RMSE	TS	RMSE	TS	RMSE	TS
mean	0.855	0.409	0.843	0.415	1.189	0.346	1.061	0.289	0.892	0.407
std	0.123	0.0395	0.109	0.0375	0.133	0.0198	0.110	0.0194	0.104	0.0425
min	0.64	0.34	0.65	0.32	0.98	0.32	0.89	0.26	0.72	0.32
median	0.84	0.42	0.84	0.42	1.19	0.35	1.07	0.29	0.88	0.42
max	1.13	0.46	1.08	0.47	1.45	0.38	1.29	0.33	1.11	0.46

TABLE 2. Comparison of model performance by MAE and BIAS

	LSTM		CNN+LSTM		Persistence		Moving-Average		ST-GNN	
	MAE	BIAS	MAE	BIAS	MAE	BIAS	MAE	BIAS	MAE	BIAS
mean	0.193	-0.0140	0.172	-0.030	0.184	0.000	0.194	0.000	0.185	0.0241
std	0.0272	0.0276	0.0290	0.0112	0.0260	0.000	0.0272	0.000	0.0237	0.0193
min	0.16	-0.082	0.14	-0.062	0.15	0.00	0.16	0.00	0.16	0.0052
median	0.17	-0.018	0.16	-0.031	0.17	0.00	0.18	0.00	0.17	0.022
max	0.25	0.037	0.24	-0.018	0.24	0.00	0.24	0.00	0.23	0.070

TABLE 3. Comparison by observation points

	LSTM				CNN+LSTM			
Observation points	RMSE	MAE	BIAS	TS	RMSE	MAE	BIAS	TS
Maebashi	0.94	0.17	-0.027	0.45	0.84	0.16	-0.035	0.45
Isesaki	0.93	0.16	-0.016	0.46	0.90	0.15	-0.025	0.44
Kamisatomi	1.13	0.22	-0.0086	0.43	1.08	0.18	-0.062	0.47
Kanra	0.84	0.17	-0.018	0.35	0.83	0.16	-0.031	0.32
Kiryu	0.82	0.19	0.0060	0.44	0.86	0.15	-0.031	0.42
Kusatsu	0.97	0.25	-0.025	0.39	0.95	0.24	-0.020	0.43
Minakami	0.83	0.22	-0.025	0.34	0.82	0.20	-0.034	0.41
Nakanojyo	0.75	0.17	0.037	0.46	0.73	0.16	-0.019	0.44
Nishinomaki	0.81	0.20	0.024	0.42	0.81	0.16	-0.033	0.41
Numata	0.68	0.17	-0.0015	0.36	0.65	0.15	-0.019	0.40
Tashiro	0.89	0.17	-0.082	0.39	0.87	0.16	-0.034	0.43
Tatebayashi	0.89	0.17	-0.021	0.43	0.91	0.14	-0.037	0.42
Fujiwara	0.64	0.23	-0.026	0.40	0.71	0.22	-0.018	0.36

In addition, Table 1 shows that the mean RMSE of the ST-GNN is higher than that of the CNN+LSTM hybrid model, whereas its mean TS is lower. Similarly, Table 2 shows that the mean MAE of the ST-GNN is higher than that of the CNN+LSTM hybrid model. These results indicate that the proposed CNN+LSTM hybrid model provides competitive performance not only against simple baseline methods but also against the graph-based reference model used in this study. Table 3 shows the evaluation indices for each observation point at a one-hour lead time. Table 3 shows that the MAE of the CNN+LSTM hybrid model is smaller than that of the standalone LSTM model for all stations. Table 3 also shows that for Maebashi, Isesaki, Kamisatomi, Kanra, Kusatsu, Minakami, Nakanojyo, Numata, and Tashiro, the RMSE of the CNN+LSTM hybrid model is smaller than that of the standalone LSTM model. Therefore, the prediction accuracy is improved at these stations. In contrast, the RMSE of the CNN+LSTM hybrid model is larger than that of the standalone LSTM model at Kiryu, Tatebayashi, and Fujiwara. The TS of the CNN+LSTM hybrid model is higher than that of the standalone LSTM model at Kamisatomi, Kusatsu, Minakami, Numata, and Tashiro. In contrast, the TS is lower at Isesaki, Kanra, Kiryu, Nakanojyo, Nishinomaki, Tatebayashi, and Fujiwara. Kiryu, Tatebayashi, and Fujiwara are shown in yellow in Figure 2. Figure 2 shows that there are only a few stations around Kiryu, Tatebayashi, and Fujiwara. In addition, Figure 2 shows that, except for Nakanojyo, the observation stations where the TS value became lower than that of the standalone LSTM model are located near the outer boundary of the prefecture. Therefore, the decrease in forecast accuracy may be attributed to the lack of observation stations in surrounding areas. In addition to evaluating the

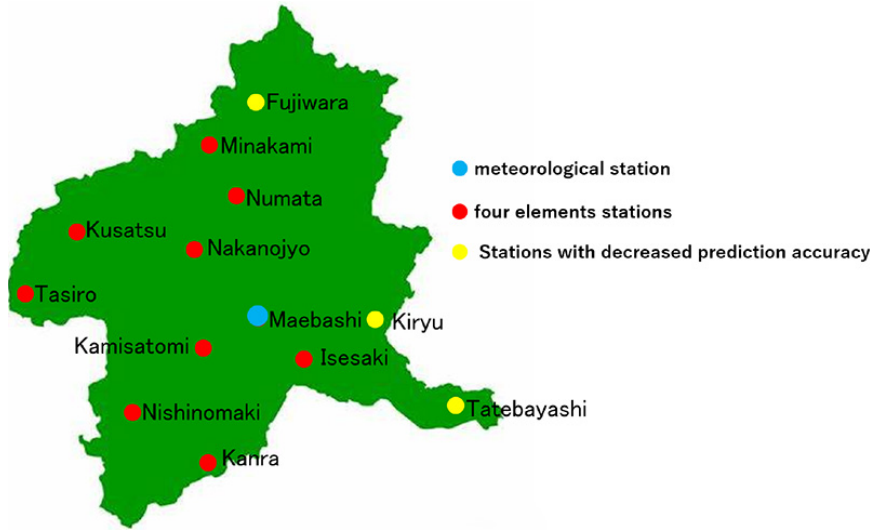


FIGURE 2. Observation points with reduced accuracy

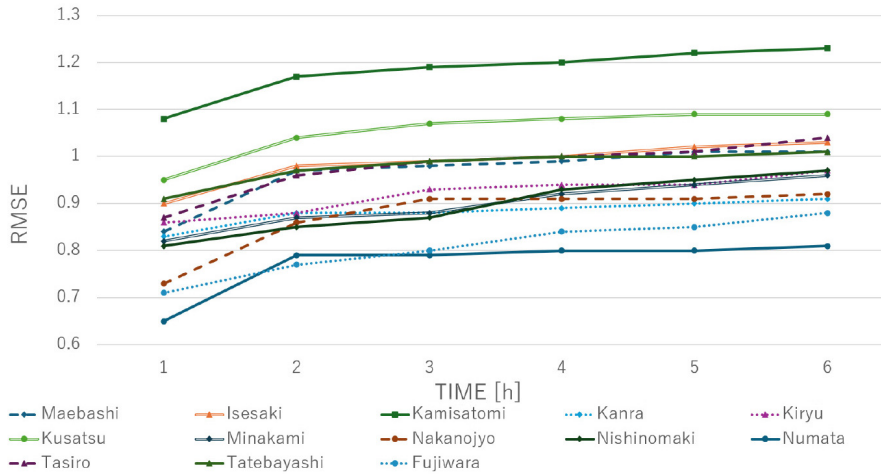


FIGURE 3. RMSE for observation points

accuracy for 1-hour-ahead forecasts, we also compared the results at 2, 3, 4, 5, and 6-hour lead times. Figure 3 shows how the RMSE of the CNN+LSTM hybrid model changes with the forecast lead time. The RMSE values for most observation stations increase the most between the 1-hour and 2-hour lead times. Figure 4 shows how the MAE of the CNN+LSTM hybrid model changes with the forecast lead time. Figure 5 shows how the BIAS of the CNN+LSTM hybrid model changes with the forecast lead time. Figure 5 shows that the BIAS at each observation point fluctuates across lead times and remains mostly around zero without a clear monotonic trend. Figure 6 shows how the TS of the CNN+LSTM hybrid model changes with the forecast lead time. The TS decreases sharply from the 1-hour to the 2-hour lead time, and then the changes become more gradual for longer lead times. Next, we present how the average values of the evaluation metrics across all the observation stations change with the forecast lead time. Figure 7 shows how the average RMSE changes with the forecast lead time. From Figure 7, the average RMSE exhibits the largest change between the 1-hour and 2-hour lead times. Figure 8 shows how the average MAE changes with the forecast lead time. Figure 8 shows that the average MAE exhibits the largest change between the 1-hour and 2-hour lead times. Figure 9

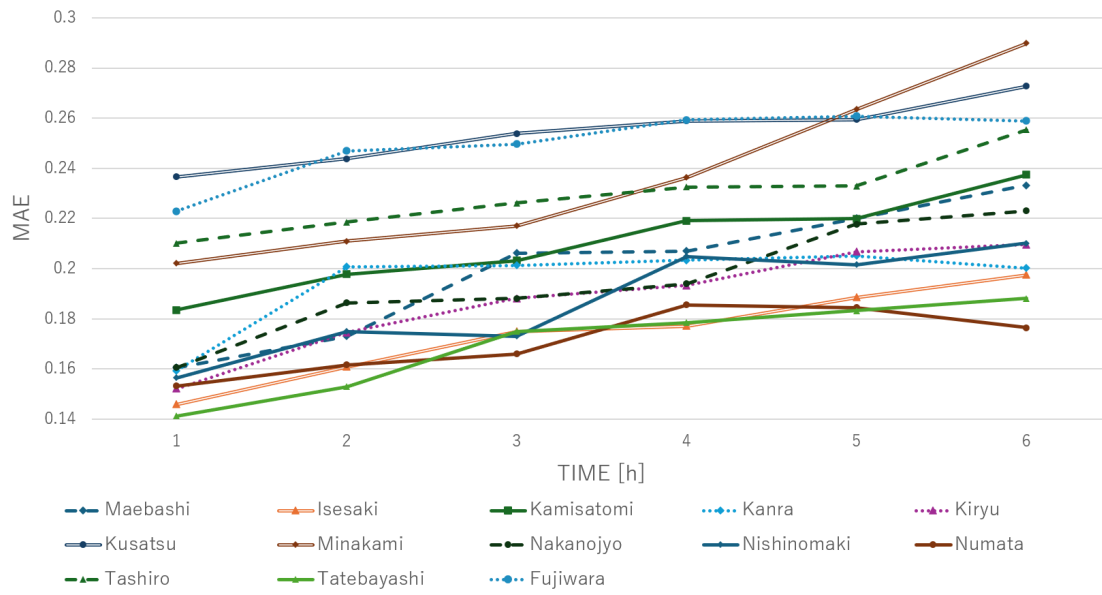


FIGURE 4. MAE for observation points

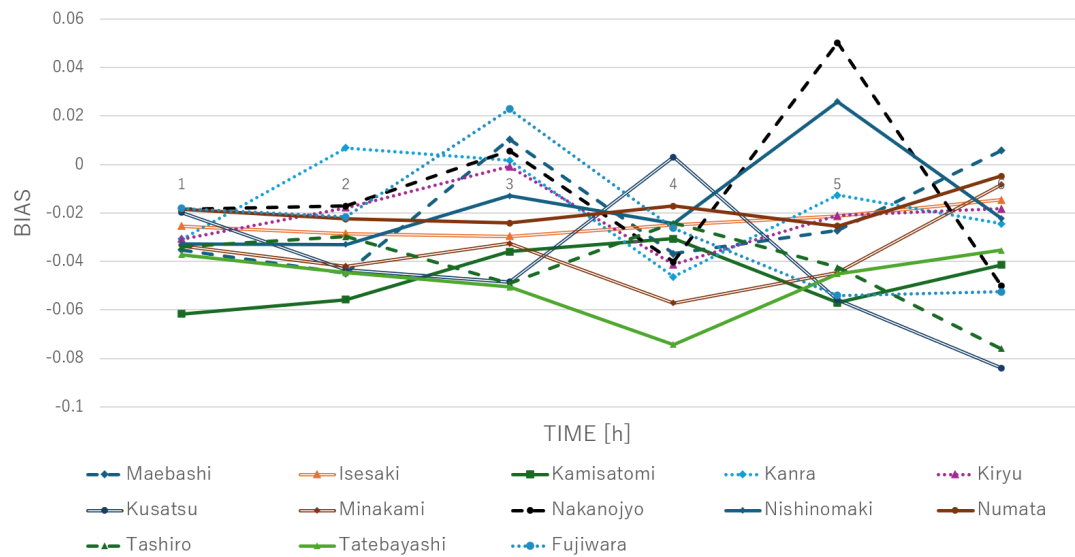


FIGURE 5. BIAS for observation points

shows how the average BIAS changes with the forecast lead time. Figure 9 indicates that the average BIAS remains close to zero and does not show a monotonic increasing or decreasing trend across lead times. Figure 10 shows how the average TS changes with the forecast lead time. From Figure 10, the average TS exhibits the largest change between the 1-hour and 2-hour lead times. The evaluation reveals that the proposed CNN+LSTM hybrid model performs best at 1-hour-ahead predictions, with diminishing improvements at longer forecast horizons. This trend supports the conclusion that our method is most effective for very short-term quantitative precipitation forecasting.

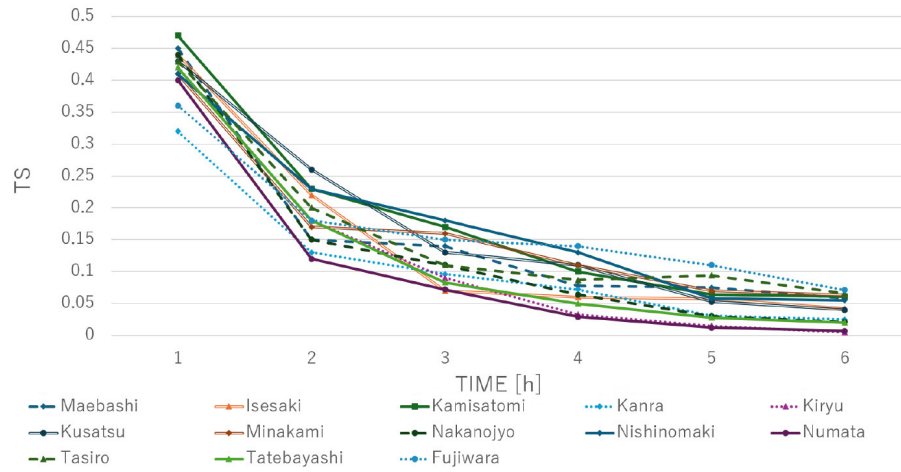


FIGURE 6. TS for observation points

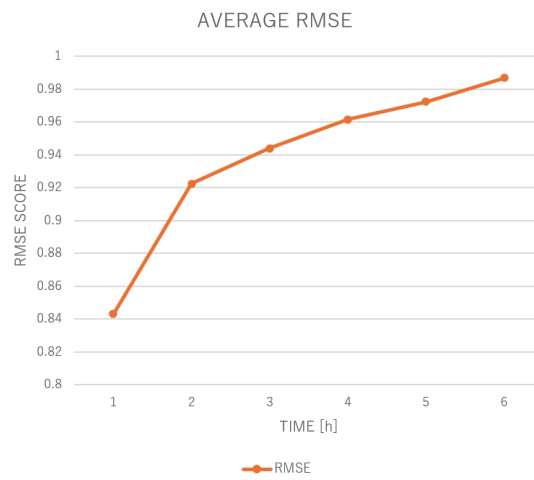


FIGURE 7. Average RMSE score

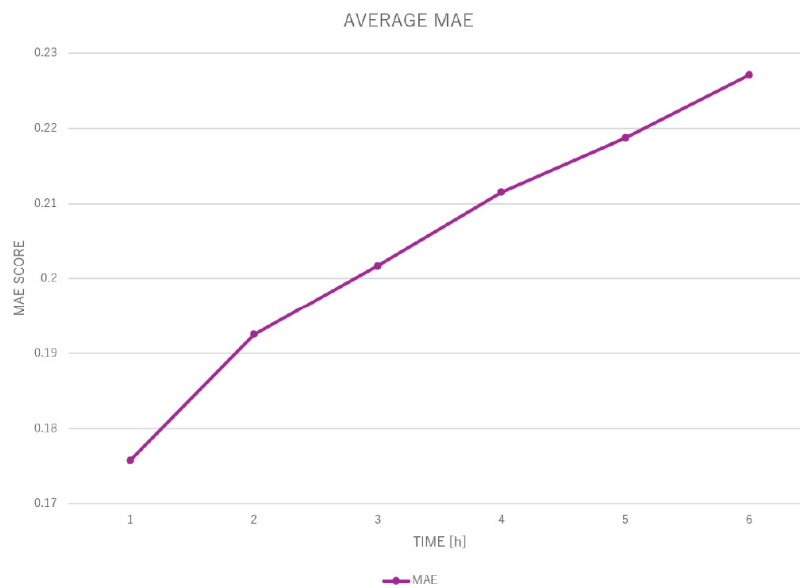


FIGURE 8. Average MAE score

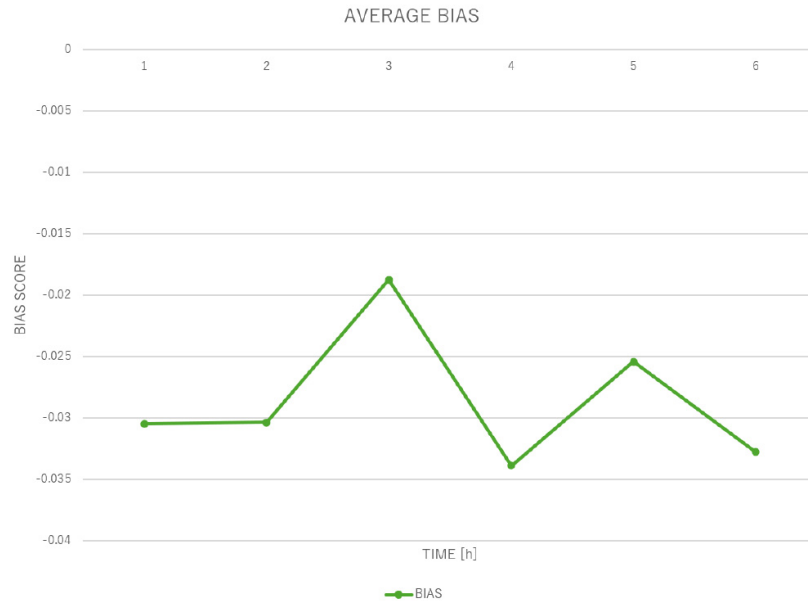


FIGURE 9. Average BIAS score

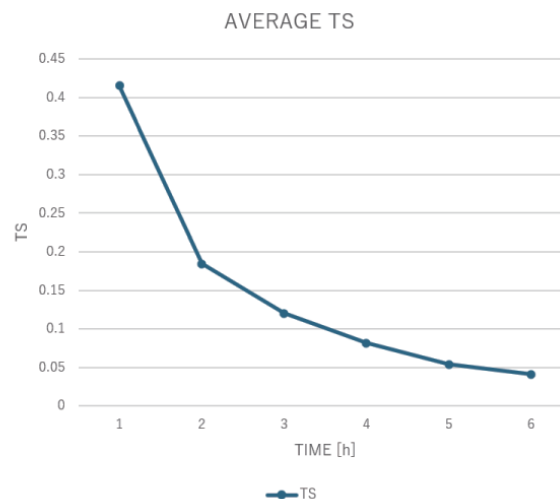


FIGURE 10. Average TS

6. Conclusions. In this study, we propose a CNN+LSTM hybrid model for short-term quantitative precipitation forecasting using observations from 13 stations in Gunma Prefecture. The main contribution of this study is to incorporate station location information through a CNN-based spatial representation and combine it with temporal modeling using LSTM. In our experiments, the proposed model achieved better performance than the standalone LSTM model and simple baseline methods for stations surrounded by densely located neighbors. These results suggest that explicitly introducing spatial context is beneficial when sufficient neighboring observations are available. However, the average improvements over the standalone LSTM model were modest, and we did not conduct statistical significance testing. Therefore, our current results should be interpreted as evidence of their potential usefulness. From a practical viewpoint, the observed dependence on station density indicates a limitation in the current spatial representation. The benefit of spatial modeling may diminish in regions with sparse stations. This finding suggests

an important scientific implication: the value of spatial information in station-based forecasting depends strongly on the availability and arrangement of neighboring observations. In this paper, we demonstrate that the proposed CNN+LSTM approach can improve short-term precipitation forecasting. However, it is necessary to further examine whether these observed improvements are statistically robust. Moreover, the comparative study should be expanded to include other spatiotemporal architectures such as attention-based and transformer-based models. Further improvement of the spatial representation is required by incorporating additional geographic information such as elevation. In addition, spatial coverage should be enhanced using additional observation points or auxiliary data sources. These problems will be addressed in future studies to clarify the conditions under which the proposed CNN+LSTM approach provides meaningful gains and to improve the robustness and generalization performance of short-term precipitation forecasting.

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